BATHINI RAJA RAJESHWAR

Hyderabad, Telangana, India | +91 9581886706 | rajarajeshwar07@gmail.com

PROFESSIONAL SUMMARY

Computer Science Engineering graduate specializing in Artificial Intelligence and Machine Learning, with hands-on experience designing and building AI-driven robotics, NLP, and software applications. Skilled in Python, Java, and machine learning fundamentals, with a track record of translating academic projects into working, end-to-end technical solutions. Strong analytical, communication, and collaboration skills; Seeking an entry-level Software Engineering / AI-ML role to contribute and grow my expertise.

Programming Languages: Python, Java, C

Web Development: HTML, CSS, JavaScript

AI/ML: Machine Learning Fundamentals, Natural Language Processing (NLP), Model Training & Evaluation

Tools & Platforms: Git, GitHub, MySQL, ROS2, Raspberry Pi

Professional Skills: Problem Solving, Communication, Team Collaboration, Time Management, Attention to Detail, Adaptability, Documentation, Quick Learner

EDUCATIONAL QUALIFICATIONS

B.Tech in Computer Science Engineering (AI & ML) — GPA: 6.46/10
JNTUH University College of Engineering, Sultanpur, Sangareddy, India | 2022 – 2026

Intermediate (MPC), Telangana State Board of Intermediate Education — 74.6%
New Chaitanya Junior College, Hyderabad, India | 2020 – 2022

SSC, Telangana State Board of Secondary Education — CGPA: 9.7/10
Mercy's Model High School, Hyderabad, India | 2017 – 2020

PROJECTS

Mobile Autonomous Robotic System with AI-Based Voice Interaction — *Major Project (NLP, ROS2)* *April 2026*

- Engineered an AI-powered autonomous mobile robot on Raspberry Pi, integrating Speech-to-Text (Whisper) and Text-to-Speech (pyttsx3/gTTS) to enable natural voice-based human-robot interaction.
- Designed a layered perception-processing-actuation architecture, incorporating ultrasonic and IR sensors for real-time obstacle detection and avoidance with DC/servo motors controlled via an L298N motor driver.
- Integrated the OpenAI API to enable context-aware, intelligent decision-making, improving the robot's responsiveness to user voice commands.
- **Technologies:** Python, Raspberry Pi, NLP, Speech-to-Text/Text-to-Speech, OpenAI API, Ultrasonic & IR Sensors, L298N Motor Driver, Embedded Systems

Video Synopsis Generator — *Mini Project (Python, NLP)* *July 2025*

- Built an Android application that automatically generates concise summaries from video content, improving accessibility for users.
- Implemented Automatic Speech Recognition (ASR) to convert video audio into structured, analyzable text data.
- Applied extractive and abstractive text summarization techniques using NLP to condense content while preserving key information.
- Integrated Python-based machine learning models into the Android application workflow, ensuring reliable end-to-end performance.

Disease Detection System — *Minor Project (Machine Learning, HTML, CSS)* *July 2024*

- Developed a machine learning-based disease detection framework using classification techniques to predict medical outcomes from patient data.
- Performed data preprocessing, feature engineering, and algorithm optimization to improve model prediction accuracy.

ACHIEVEMENTS & CERTIFICATIONS

- Completed a two-day Workshop on Full Stack Development at JNTUHCES.
- Completed a Workshop on Android Development, Git, and GitHub at JNTUHCES.
- Competed in a 24-hour Hackathon at JNTUHCES, collaborating under time pressure to design and build a working prototype.